

4

MEASURES OF SHAPE OF A DISTRIBUTION



interpret and evaluate measures of skewness and kurtosis to address an investment problem

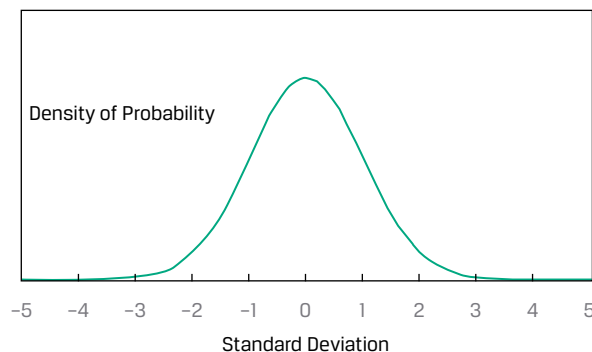
Mean and variance may not adequately describe an investment's distribution of returns. In calculations of variance, for example, the deviations around the mean are squared, so we do not know whether large deviations are likely to be positive or negative. We need to go beyond measures of central tendency, location, and dispersion to reveal other important characteristics of the distribution. One important characteristic of interest to analysts is the degree of symmetry in return distributions.

If a return distribution is symmetrical about its mean, each side of the distribution is a mirror image of the other. Thus, equal loss and gain intervals exhibit the same frequencies.

One of the most important distributions is the normal distribution, depicted in Exhibit 16. This symmetrical, bell-shaped distribution plays a central role in the mean–variance model of portfolio selection; it is also used extensively in financial risk management. The normal distribution has the following characteristics:

- Its mean, median, and mode are equal.
- It is completely described by two parameters—its mean and variance (or standard deviation).

Exhibit 16: The Normal Distribution



Other distributions may require more information than just the mean and variance to characterize their shape.

Skewness

A distribution that is not symmetrical is termed **skewed**. A return distribution with positive skew has frequent small losses and a few extreme gains. A return distribution with negative skew has frequent small gains and a few extreme losses. Panel A